https://lucid.app/lucidchart/e426cb76-a4f8-475e-a6d4-bcc0630b3848/edit?viewport_loc=421%2C943%2C2366%2C1202%2C0_0&invitationId=inv_91ca6234-643d-4dc2-80eb-c360159d061f

Relational Schema

- User(user_id, username, email, password)
- Artist(artist_id, name, genre, biography)
- Venue(venue_id, name, city, state, capacity, genreTags)
- Song(song_id, title, *artist_id*)
- Show(show_id, date, ticketPrice, genre, *venue_id*)
- Setlist(setlist_id, *show_id,* song_id, order)
- Attendance(attendance_id, *user_id, show_id*, rating, review)
- ShowArtist(*show_id, artist_id*)
- UserFollows(*user_id, artist_id*)

BCNF Proof

- User(user_id, username, email, password)

The only non-trivial FD has user_id as its determinant, which is the primary key and therefore a superkey.

- Artist(artist_id, name, genre, biography)

The only non-trivial FD has artist_id as its determinant, which is the primary key and therefore a superkey.

- Venue(venue_id, name, city, state, capacity, genreTags)

The only non-trivial FD has venue_id as its determinant, which is the primary key and therefore a superkey.

- Song(song_id, title, *artist_id*)

The only non-trivial FD has song_id as its determinant, which is the primary key and therefore a superkey.

- Show(show_id, date, ticketPrice, genre, *venue_id*)

The only non-trivial FD has show_id as its determinant, which is the primary key and therefore a superkey

- Setlist(setlist_id, *show_id,* song_id, order)

The only non-trivial FD has setlist_id as its determinant, which is the primary key and therefore a superkey.

- Attendance(attendance_id, *user_id, show_id*, rating, review)

The only non-trivial FD has attendance_id as its determinant, which is the primary key and therefore a superkey.

- ShowArtist(*show_id, artist_id*)

There are no non-trivial functional dependencies in this relation because it contains only key attributes. A relation with no non-trivial FDs trivially satisfies BCNF

- UserFollows(*user_id, artist_id*)

There are no non-trivial functional dependencies in this relation because it contains only key attributes. A relation with no non-trivial FDs trivially satisfies BCNF.